Emotion Analysis System Using Deep Learning

Project Summary

Introduction

The Emotion Analysis System is a desktop application developed to recognize human emotions from facial expressions using deep learning and computer vision techniques. The application analyzes facial images captured from a live webcam or an uploaded image and predicts the emotion displayed on the person's face. The main objective of this project is to demonstrate how artificial intelligence can be used to understand facial expressions and classify emotions in real time.

Objective

The objective of this project is to develop a facial emotion recognition system that can accurately identify emotions such as **Happy, Sad, Angry, Neutral, and Surprise**. The application also displays the confidence score of the prediction, helping users understand how confident the model is in its result.

Dataset

The model was trained using the **FER2013 (Facial Expression Recognition 2013)** dataset. This dataset contains approximately 35,000 grayscale facial images of size **48 × 48 pixels**. The images are categorized into seven emotion classes: Angry, Disgust, Fear, Happy, Sad, Surprise, and Neutral. During training, the CNN model learned important facial features from these images to classify emotions accurately.

Methodology

The development of this project started with preparing the FER2013 dataset. All images were resized to 48 × 48 pixels and normalized before training. Data augmentation techniques such as rotation, zooming, and horizontal flipping were applied to improve the performance and generalization of the model.

A Convolutional Neural Network (CNN) was built using TensorFlow and Keras. The network consists of convolutional layers, pooling layers, dropout layers, and fully connected layers. These layers automatically learn facial features such as the eyes, eyebrows, nose, and mouth, which are useful for identifying different emotions.

After training, the model was saved and used for emotion prediction. OpenCV was used to access the webcam and perform face detection using a Haar Cascade classifier. Each detected face is converted to grayscale, resized to 48 × 48 pixels, normalized, and passed to the trained CNN model. The model predicts the emotion and calculates the confidence score. The detected emotion and confidence percentage are displayed above each detected face in real time.

The application also supports uploaded images. When an image is provided, the system detects all faces present in the image, predicts the emotion for each face individually, and displays the corresponding confidence score. During webcam analysis, the detected emotions are recorded, and an emotion trend graph is generated after the session to summarize the frequency of each detected emotion.

Technologies Used

The following technologies were used in this project:

- Python
- TensorFlow
- Keras
- OpenCV
- NumPy
- Matplotlib

Features

The developed application provides the following features:

- Emotion detection from uploaded images.
- Real-time emotion recognition using a webcam.
- Multiple face detection.
- Emotion prediction with confidence score.
- Emotion trend visualization after webcam analysis.

Applications

Emotion recognition has several practical applications, including:

- Human-computer interaction.
- Mental health monitoring.
- Smart classroom systems.
- Customer behavior analysis.
- Healthcare assistance.
- Driver monitoring systems.

Conclusion

The Emotion Analysis System successfully demonstrates the use of deep learning and computer vision for real-time facial emotion recognition. By combining TensorFlow for emotion classification and OpenCV for face detection and webcam access, the application accurately predicts facial emotions and displays confidence scores. The project meets all the required objectives, including image-based emotion recognition, webcam-based real-time detection, multiple face detection, and emotion trend analysis. It also provides a strong foundation for future improvements such as using more advanced deep learning models and deploying the application on cloud or mobile platforms.